

Layered Certificates for Static Cost Forecasting in Pauli Compilation

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Abstract

Static cost formulas are useful only if their authority is stated precisely: benchmark agreement, theorem-backed exactness, and a compact structural explanation are different claims. We make that separation explicit for a three-block shared-tag Pauli compilation grammar with a unit structural support-count objective. An interpretable three-family formula matched 9,545 of 9,546 distinct benchmark instances and all 102 predictions staged before exact evaluation, yet one three-qubit instance refuted its equality claim, with exact cost 10 and formula value 11. We retain that adverse row and replace the formula with an exact support-at-most-two evaluator. A support-reduction theorem proves that this evaluator equals the unrestricted optimum for every qubit count in scope; zero error over 9,547 comparison events then checks the implementation rather than serving as the proof. A first enlargement of the explanation family repaired the original row but failed on 64 of 740 hostile cases. A hybrid family covered 10,481 verified finite instances, and a complete local-domain argument subsequently closed the explanation classification for all sizes under the unit objective. Two failed proof routes are retained because they mark genuine limits of simpler arguments. Under a pre-specified reweighted objective the support theorem does not transfer: an exact witness has costs $11 < 13 < 23$ for the unrestricted, support-two, and weight-one families. The resulting certificate reports feasible bounds, theorem-backed cost, explanation, and row-level validation separately; it makes no claim about full-circuit, fault-tolerant, or hardware resource optimality.

Keywords: quantum compilation; Pauli operators; static resource analysis; exact optimization; machine-checked proof; falsification.

1 Introduction

Reliable quantum resource analysis spans several levels. Full-stack estimators translate algorithms and error-correction assumptions into physical resource budgets [1]; language-level analyses derive costs or width from program structure [2, 3, 4]; and verified optimizers establish semantic preservation for circuit rewrites [5, 6]. Between these levels lies a narrower but common compiler question: can a structural cost be predicted before running an unrestricted optimizer, and what exactly certifies the prediction?

A low error rate is not an exactness theorem. A theorem about an exact evaluator does not, by itself, provide a compact explanation of why a particular input is cheap. A feasible construction provides an upper bound, but not equality. Conflating these statements is especially risky when a forecast is used downstream as if it were a measured resource. We therefore treat a static forecast as a stack of independently testable certificates.

We study a three-block shared-tag grammar for compiling six Pauli targets. The objective counts structural supports with fixed slot multipliers and a local restoration charge. Our scope is deliberately narrow: the target grammar is fixed, Pauli phases are quotiented out, and the main theorems concern only the unit support-count objective defined in Sec. 2. They do not estimate logical error, physical qubits, wall-clock execution, or fault-tolerant overhead.

The main results are:

1. **A retained exact refutation.** An interpretable three-family formula survives an exhaustive two-qubit slice and a prospectively staged panel, but one frozen three-qubit row gives exact cost 10 and formula value 11. The tuple of target pairs is

$$((XYZ, YYX), (YYX, XXZ), (ZZI, IYI)).$$

The adverse row is part of the result, not an outlier removed during model repair.

2. **All-size exactness of a support-two evaluator.** A support-reduction argument proves that every unrestricted optimum has a representative whose frame strings have support at most two. The resulting evaluator is exact for every qubit count within the fixed grammar and objective. Agreement over 9,547 comparison events checks the implementation; it is not substituted for the proof.
3. **A separately falsifiable explanation layer.** The first repaired explanation is refuted by 64 hostile cases. A hybrid family covers 10,481 verified finite instances, after which a complete local-domain domination argument closes the explanation classification for all sizes. The cost certificate remains valid throughout this sequence.
4. **Explicit non-transfer.** A pre-specified reweighting admits an unrestricted cost of 11 while the support-two and weight-one restrictions cost 13 and 23. The unit-objective theorem therefore cannot be exported by analogy.

This separation changes how negative evidence is interpreted. The initial formula failure localizes a missing explanation family. The later failure of an enlarged family shows that explanation still lagged exact prediction. Neither failure contradicts support reduction. Conversely, support reduction cannot rescue a changed objective for which its local exchange inequality is false.

Section 2 defines the grammar and objective. Section 2.2 describes the certificate and evidence design. Section 3 presents the theorem, counterexamples, and explanation closure. We then position the result among static resource analysis, verified compilation, and Pauli-aware optimization, before stating limitations and reproducibility requirements.

2 Compilation setting and certificate design

2.1 Pauli representation and feasible configurations

We work with n -qubit Pauli strings modulo global phase. A string is represented by a binary symplectic key $(x, z) \in \mathbb{F}_2^{2n}$; its support is the set of qubits on which $(x_q, z_q) \neq (0, 0)$, and $\text{wt}(\cdot)$ denotes support size. Two strings anticommute when their binary symplectic inner product is one.

An input t contains three blocks indexed by $j \in \{0, 1, 2\}$, each with two Pauli targets $P_{j,0}$ and $P_{j,1}$. A feasible compilation configuration u selects:

- frame strings $R_{j,0}$ and $R_{j,1}$ that anticommute within every block;

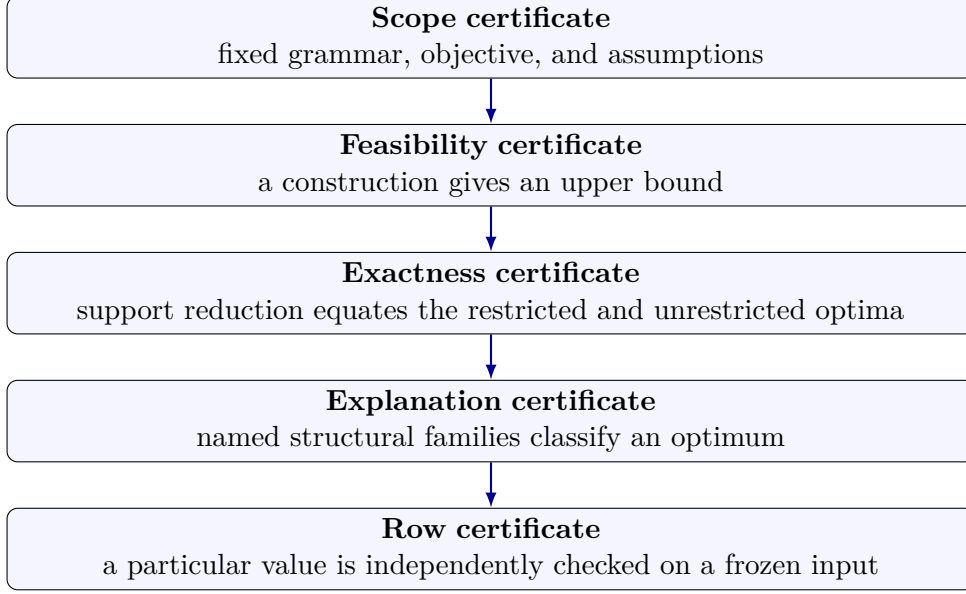


Figure 1: The certificate layers answer different questions. A failure below one layer does not automatically falsify a stronger statement above it; for example, an incomplete explanation family need not invalidate a theorem-backed exact evaluator.

- one shared tag string S ;
- a matching, a relative assignment of the two targets to the two frames in each block, and one central-slot choice per block.

The restoration strings are $T_{j,k} = P_{j,k}R_{j,k}$, with Pauli multiplication taken modulo phase.

For three single-qubit Pauli letters, define

$$\phi(a, b, c) = \begin{cases} 1, & a = b = c \neq I, \\ \mathbf{1}[a \neq I] + \mathbf{1}[b \neq I] + \mathbf{1}[c \neq I], & \text{otherwise.} \end{cases} \quad (1)$$

The unit structural support-count cost is

$$c(u; t) = \sum_{j=0}^2 \sum_{k=0}^1 m_{j,k} \text{wt}(R_{j,k}) + 2 \text{wt}(S) - 18 + \sum_{k=0}^1 \sum_{q=1}^n \phi(T_{0,k,q}, T_{1,k,q}, T_{2,k,q}), \quad (2)$$

where $m_{j,k} = 2$ for a central slot and $m_{j,k} = 4$ otherwise. The constant fixes the same zero point for all feasible choices and has no effect on the minimizer.

The unrestricted exact optimum and the support-at-most-two optimum are

$$C_{\star}(t) = \min_{u \in \mathcal{U}(t)} c(u; t), \quad (3)$$

$$C_{\leq 2}(t) = \min_{u \in \mathcal{U}(t): \max_{j,k} \text{wt}(R_{j,k}) \leq 2} c(u; t). \quad (4)$$

The equality $C_{\star} = C_{\leq 2}$ is a theorem only for Eq. (2); Sec. 3.5 gives a counterexample to transfer under reweighting.

2.2 Forecast and explanation families

We use descriptive names for restricted feasible families. The *common-anchor* family shares a simple anchor pattern across blocks. The *weight-one arbitrary-anchor* family permits independent single-support anchors. The *restricted-borrow* family permits a central block to borrow a restoration location subject to its original support rule. Their initial explanatory forecast is

$$F_{\text{simple}}(t) = \min\{C_{\text{ca}}(t), C_{\text{wa}}(t), C_{\text{b}}(t)\}. \quad (5)$$

Every term is a feasible-family minimum, so $C_{\star} \leq F_{\text{simple}}$; equality was initially an empirical hypothesis, not a theorem.

The exact forecast is instead

$$F_{\text{exact}}(t) = C_{\leq 2}(t). \quad (6)$$

It is evaluated by an enumerator restricted to full frame support at most two. The forecast path does not call the unrestricted dynamic program. Exactness follows from the support-reduction theorem in Sec. 3.1; finite comparisons check that the implementation respects the theorem.

For interpretation, we also study an *enlarged-borrow* family, which allows a borrow home outside the borrowing block’s own target support, and a *hybrid* family combining two tag anchors with a phantom borrow. Their costs are denoted $C_{\text{b}+}$ and C_{h} . The final explanation theorem is

$$C_{\star} = C_{\leq 2} = \min\{C_{\text{wa}}, C_{\text{b}+}, C_{\text{h}}\} \quad \text{for the unit objective.} \quad (7)$$

These family labels explain a minimizing structure; they are not separate physical resource models.

2.3 Chronological evidence design

The evaluation was staged so that refutations could not be silently absorbed into later definitions:

1. establish exact truth for a structured two-qubit slice and fixed application-derived rows;
2. freeze predictions for a public-library panel and engineered boundary cases before exact evaluation;
3. freeze a fresh seeded panel, retain the first mismatch, and bind its exact witness;
4. introduce the theorem-backed support-two evaluator without changing the adverse row;
5. freeze hostile shapes against the first repaired explanation;
6. assess the hybrid family on the accumulated and adversarial finite domains;
7. close the remaining all-size explanation sector on a complete local state domain.

We distinguish *distinct inputs* from *comparison events*. The exact evaluator was compared 9,547 times on 9,546 distinct inputs because the refuting row was present in the seeded panel and was also rechecked under a separately bound audit. We report both numbers rather than treating the repeat as an independent instance.

No runtime comparison is retained. Runtime depends on the final implementation and execution environment; it is not part of the exactness claim.

Table 1: Exact comparison of the initial explanatory formula. The seeded panel includes the retained refuting input.

Panel	Distinct inputs	Matches	Mismatches
Structured two-qubit slice	9,261	9,261	0
Fresh seeded two-/three-qubit panel	240	239	1
Application-derived subjects	45	45	0
Total	9,546	9,545	1

3 Results

3.1 Support at most two is exact under the unit objective

Theorem 1 (Support reduction). *For every qubit count, target six-tuple, matching, relative target assignment, and central-slot choice in the grammar of Sec. 2, the unit-objective optimum is attained by a configuration in which every frame string has support at most two. Hence*

$$C_{\star}(t) = C_{\leq 2}(t).$$

The proof is local but yields an all-size statement. For a frame R with support $w \geq 3$ and anticommuting partner R' , label each support coordinate by

$$(\alpha_q, \beta_q) = (\langle R_q, R'_q \rangle, \langle S_q, R_q \rangle) \in \mathbb{F}_2^2.$$

Anticommutation makes the total α parity odd. Any such multiset of size at least three contains a nonempty proper singleton or pair whose α and β sums both vanish. Zeroing those letters therefore preserves the relevant anticommutation and tag syndrome without tag repair. A complete 18,432-case local enumeration verifies that the increase of the restoration term in Eq. (2) never exceeds the refunded frame cost; there are zero violations. Repeating the exchange decreases total frame support without increasing cost. Lexicographic descent on cost and total frame support terminates at support at most two. Appendix A gives the argument and its exact computational obligation.

Support two is a genuine boundary of this proof. At $w = 2$, four odd-parity class patterns lack a nonempty proper zero-sum subset. Thus the theorem does not follow by simply repeating a support-one argument.

3.2 Finite evidence refutes the initial formula

Table 1 separates the benchmark components. The initial formula matches the complete structured two-qubit slice and all application-derived rows. In the seeded panel it misses exactly one three-qubit input. Across the 9,546 distinct benchmark inputs, the outcome is therefore 9,545 matches and one mismatch.

The mismatch has $n = 3$ and target pairs

$$((XYZ, YYX), (YYX, XXZ), (ZZI, IYI)).$$

Exact optimization gives $C_{\star} = 10$, whereas all three terms in Eq. (5) equal 11. Thus

$$C_{\star} = 10 < 11 = F_{\text{simple}}. \quad (8)$$

Its minimizing support-two frame uses a borrow home outside its block’s own target support. The row refutes only the completeness of the three-family explanation. It does not refute feasibility of the formula, because $10 \leq 11$, or the support theorem, because $C_\star = C_{\leq 2} = 10$.

Before this mismatch was known, 102 predictions had been staged and then exactly evaluated: 90 public-library matchings and 12 engineered boundary rows, with four examples allocated to each of the three initial configurations. All 102 matched. These prospective successes provide calibration evidence, but Eq. (8) shows why they cannot establish global equality.

3.3 The exact evaluator passes its implementation checks

The support-two evaluator returns the exact cost on all 9,547 comparison events, representing 9,546 distinct inputs. It also returns 10 on the refuting row. The unrestricted optimizer is absent from its forecast path. For higher-qubit application rows, exactness is sometimes certified by the containment pinch

$$C_\star \leq C_{\leq 2} \leq C_{\text{wa}} = C_\star,$$

rather than by a direct support-two sweep. This distinction matters: zero finite error validates consistency of the code and witnesses, whereas the support-reduction proof is the source of the all-size claim.

3.4 Explanation repair, hostile refutation, and all-size closure

Allowing borrow homes outside a block’s own target support repairs Eq. (8). That repair is still incomplete. A frozen hostile search evaluates 740 exact cases and finds 64 in which the enlarged-borrow explanation remains one unit above $C_{\leq 2}$. All 64 optima stay within support two, so the exact forecast is unaffected.

Adding the hybrid family covers 10,481 verified finite inputs, including the 64 hostile witnesses, with no observed need for a fifth configuration. This finite zero-residual result is useful but is not the all-size proof. The final proof isolates the remaining commuting support-two blocks and admits independent permutations of each block’s two targets, a degree of freedom already present in the feasible configuration space. The complete domain comprises 378 irreducible geometries and 16,777,216 target-letter states per geometry. Every state has a non-increasing alternative with strictly fewer commuting support-two blocks; no residue remains. Induction on their number proves Eq. (7) for all qubit counts under the unit objective.

3.5 Adverse boundary under a changed objective

A pre-specified reweighted objective changes the local exchange economics. On one exact witness,

$$C_\star = 11 < C_{\leq 2} = 13 < C_{\text{wa}} = 23. \tag{9}$$

The corresponding panel contains 53 failures of support-two closure. Equation (9) is not a perturbative degradation of the main theorem; it falsifies transfer of the theorem to that objective. Any new weighting requires a fresh exchange inequality or a different structural argument.

4 Discussion

The central methodological result is not merely that one restricted search happens to be exact. It is that feasibility, exact cost, structural explanation, and finite row verification can be assigned

Table 2: Certificate chronology. A finite check is not promoted into the proof column.

Layer	Exact evidence	Conclusion
Initial explanation	9,545/9,546 distinct inputs; 102/102 prospectively staged rows	Refuted by one exact row
Exact forecast	Support-reduction theorem; 9,547 zero-error comparison events	Exact for all sizes in scope
Enlarged borrow	64 failures among 740 hostile cases	Explanation refuted; cost theorem intact
Hybrid explanation	10,481 finite inputs with no residual	Finite coverage only
Final explanation	Complete domains for 378 geometries; no local residue	All-size classification under the unit objective

separate authorities. The separation makes a counterexample diagnostic. The first adverse row identified a missing borrow geometry. The hostile panel then showed that the first repair was also incomplete. In both cases the support-two evaluator remained exact because its certificate did not depend on the explanation family.

This design also prevents a common inversion of evidence. The 9,547 zero-error comparison events are strong implementation evidence, but they cannot prove behavior for arbitrary qubit count. The proof reduces the all-size claim to a combinatorial subset lemma and a finite local inequality. Conversely, the proof does not certify that a particular executable, serialization, or table was produced correctly; those are checked by independent witnesses and arithmetic consistency tests.

The explanation theorem adds value beyond exact cost. A family label exposes whether the minimizer uses independent weight-one anchors, an out-of-support borrow, or a two-anchor hybrid. Such labels can guide compiler diagnostics and future simplification, but they should not be confused with physical mechanisms. They classify feasible configurations within the abstract grammar of Eq. (2).

Negative proof routes are scientifically useful here. A replacement-orbit argument failed because its ten candidate images returned to the same qubit pair and provided no strict descent. A separate single-pinner normalization survived a complete 133,349,376-configuration product domain, yet its proposed global chain composition remained unproved. A later exact three-tag configuration refuted the premise that every chain can be represented on two coordinates. These failures do not weaken the successful per-block-permutation proof; they explain why apparently simpler globalizations were not used. Appendix B records their logical status.

The objective counterexample is equally important. Equation (9) shows that “support two is enough” is not a property of the grammar alone. It results from the exact balance between frame refund and local restoration increase in the unit objective. Reweighting can make a larger support profitable. Reporting this boundary is preferable to suggesting robustness that has not been proved.

No canonical speed comparison is claimed. Restricted enumeration may be substantially cheaper than unrestricted optimization in some regimes, but runtime depends on implementation choices, caching, symmetry reductions, and hardware. Correctness and speed should be benchmarked separately on the final public implementation.

5 Related work and residual contribution

Static quantum cost analysis has been developed at several abstraction levels. Quantum expectation transformers provide compositional cost reasoning for probabilistic quantum programs [2]. Type systems have been used to infer flexible resource bounds and circuit width [3, 4], while control-flow analyses expose the non-Clifford cost of error correction [7]. Full-stack estimators instead connect logical programs, error-correction choices, and hardware assumptions [1]. These approaches motivate certified and early-stage resource reasoning, but our cost is a compiler-specific structural objective rather than a general program or hardware resource vector.

A second line of work verifies circuit optimization. The verified optimizer VOQC proves semantics-preserving rewrites in a proof assistant [5]; Giallar automates verification conditions for compiler passes [6]. Our theorem certifies the value of a restricted structural optimization, not semantic equivalence of an arbitrary circuit transformation. The two forms of assurance are complementary.

Pauli structure is central to simulation compilers and block-encoding methods. Paulihedral performs block-wise optimization of Pauli-string kernels [8]; PCOAST uses a Pauli intermediate representation [9]; global binary-symplectic simplification has recently been proposed for Hamiltonian simulation [10]; and stabilizer-formalism constructions support block encodings of Pauli linear combinations [11]. We adopt the neighboring algebraic language but do not claim these compiler or block-encoding mechanisms as contributions.

The residual contribution is the combination of (i) an all-size support-reduction certificate for the exact structural cost, (ii) a separately falsifiable explanation classification, (iii) chronologically staged positive and hostile tests, and (iv) row-level distinction between theorem authority and executable verification. A bounded literature search ending on 28 August 2026 did not locate an equivalent combination for this grammar. That search is not a novelty certificate, and we make no claim of being the first static resource analysis, verified compiler, or Pauli-aware optimizer.

6 Limitations

Grammar and objective. Theorems 1 and Eq. (7) apply only to the three-block shared-tag grammar and the unit structural support-count objective in Eq. (2). They do not transfer to different tag ranks, numbers of blocks, rotation trade-offs, or coefficient-weighted costs. Equation (9) is an explicit adverse boundary.

Resource meaning. The predicted value is an abstract compilation cost. It is not a count of physical qubits, logical qubits, magic states, gates, energy, or elapsed time. It does not establish quantum advantage or deployment readiness.

Finite panels. The prospective, application-derived, seeded, and hostile panels validate executable behavior only on their stated rows. Application-derived higher-qubit equalities may use exact containment rather than direct restricted enumeration. The all-size claims come from the support and classification proofs, not from these panels.

Machine-checked local obligations. The support proof uses an exhaustive 18,432-case local inequality, and the explanation proof uses a complete 2^{24} target-letter domain for each of 378 geometries. The accompanying checker verifies the published aggregate arithmetic and selected witness relations; reproducing the complete searches requires the full scientific implementation and a suitable compute environment.

Performance. We omit historical timings and claim no asymptotic or empirical speedup. A canonical comparison should be performed only after the public implementation and execution

environment are fixed.

Literature and external authority. The literature search was bounded rather than exhaustive. Neither internal exact checks nor a manuscript build substitutes for independent peer review. A persistent public archive, license choice, and final preprint deposit remain publication actions to be completed by the author.

7 Conclusion

An interpretable static formula can be useful and still be wrong as a universal equality. Here, a single retained three-qubit counterexample separates that formula from a theorem-backed support-two evaluator. Later hostile cases separate exact cost from the first repaired explanation. Complete local-domain domination then closes the explanation classification without erasing either failed model.

The result is a layered certificate for one Pauli compilation grammar: feasible constructions bound the cost, support reduction certifies exactness, named families explain a minimizer, and frozen comparisons validate particular executable rows. The changed-objective witness shows exactly where this authority stops. Preserving these distinctions makes static forecasts more auditable and makes counterexamples informative rather than inconvenient.

8 Data, code, and reproducibility

All numerical claims arise from finite algebraic enumeration or exact optimization over Pauli keys; no human-subject, clinical, or personal data are used. Sanitized supplementary review material accompanying this manuscript contains:

- a machine-readable evidence summary with the benchmark decomposition, adverse rows, comparison-event accounting, hostile-search counts, complete-domain counts, and changed-objective boundary;
- a deterministic checker for the published arithmetic and inequality relations;
- a reviewer guide that distinguishes all-size proof obligations from finite validation.

The full scientific implementation and detailed execution receipts are available to editors and referees during review. They are kept separate from the reader-facing archive because the latter contains only neutral scientific names and no development-history metadata. Before public submission, the author will deposit a persistent, licensed snapshot and bind its identifier to the final manuscript version. Until that deposit exists, no permanent-public-archive claim is made.

A faithful reproduction should preserve chronology: evaluate the initial formula, retain its mismatch, check the support-two evaluator on the same inputs, test the enlarged explanation on the frozen hostile panel, and only then apply the final classification theorem. Collapsing the sequence into a final zero-error table would remove the falsification evidence that motivates the certificate design.

9 Declarations

Ethics. This study uses no human participants, animals, clinical records, or personal data. The principal integrity risk is epistemic over-certification; the manuscript addresses it by separating proof, finite validation, explanation, and row status.

Author contributions. Sze Chun Yiu: conceptualization, methodology, software, formal analysis, investigation, validation, data curation, writing—original draft, writing—review and editing, visualization, supervision, and project administration.

Use of generative artificial intelligence. ChatGPT was used to assist with manuscript organization, language refinement, and consistency checks. The author reviewed the resulting text, claims, equations, citations, and evidence boundaries and remains responsible for the work.

A Support-reduction proof

We expand the two finite obligations used in Theorem 1. Let R be a frame with anticommuting partner R' and support W . For each $q \in W$, write

$$\gamma_q = (\alpha_q, \beta_q) = (\langle R_q, R'_q \rangle, \langle S_q, R_q \rangle) \in \mathbb{F}_2^2.$$

The first component sums to one because R and R' anticommute.

Lemma 1 (Proper zero-sum subset). *If $|W| \geq 3$ and $\sum_{q \in W} \alpha_q = 1$, then there is a nonempty proper subset $Q \subset W$ with $|Q| \leq 2$ and $\sum_{q \in Q} \gamma_q = (0, 0)$.*

Proof. If a coordinate has class $(0, 0)$, take that singleton. Otherwise, if a class repeats, take two coordinates of that class, whose sum vanishes over $\mathbb{F}_2^2$. If neither occurs, all classes are distinct and nonzero. There are only three nonzero classes. Thus $|W| = 3$ and the classes are $(0, 1)$, $(1, 0)$, $(1, 1)$, whose first components sum to zero, contradicting the odd-parity assumption. $\square$

Zero the letters of R on Q . The zero sums preserve both the anticommutation bit against R' and the tag syndrome, while properness keeps R nonidentity. Only the chosen coordinates can change the cost. The relevant local state consists of the class (α, β) , the central or noncentral multiplier, the current restoration letters in the three blocks, and the updated letters after zeroing. Complete enumeration gives 18,432 cases. In every case,

$$\Delta \left(\sum_{k,q} \phi \right) \leq \text{refunded frame-support cost};$$

there are zero violations. The maximum net change is zero, so the exchange never increases Eq. (2).

Choose an unrestricted optimum with minimum total frame support. If any frame has support at least three, the lemma and local inequality produce another feasible configuration with no higher cost and strictly smaller total support, a contradiction. Therefore an optimum exists with every frame supported on at most two qubits.

For $|W| = 2$, the subset lemma has four failing odd-parity patterns. This is why the argument stops at support two rather than support one; the boundary is structural, not a computational cutoff.

B Explanation-closure audit

After support reduction, normalization handles the noncommuting support-two shapes and leaves configurations containing commuting support-two blocks. The successful final step treats each remaining block's assignment of its two targets to its two frames as an independent feasible permutation. For each of 378 irreducible geometries, the complete target-letter domain at the two active

qubits contains $2^{24} = 16,777,216$ states. Every state admits an alternative with cost change $\Delta \leq 0$ and strictly fewer commuting support-two blocks. Induction on the count of such blocks reduces every optimum to the union of the weight-one arbitrary-anchor, enlarged-borrow, and hybrid families. This proves Eq. (7).

The following attempted routes remain relevant because their failure prevents a misleadingly short proof.

Table 3: Retained negative proof evidence.

Attempt	Exact outcome	Consequence
Replacement-orbit descent	Ten candidate images re-entered the same qubit pair; the best available changes tied at zero and no strict global descent was obtained.	The orbit argument does not close the sector.
Single-pinner normalization	Confirmed over 133,349,376 composite configurations, but composition into a global chain was not proved.	A valid local lemma is not a global theorem.
Two-coordinate chain representation	Refuted by an exact three-tag configuration with multiple irreducible commuting support-two blocks; no new minimizing family was found.	That proof premise fails, while the per-block-permutation theorem remains intact.

These outcomes are compatible: a local normalization can be correct, its proposed chain representation can fail, and a different complete-domain induction can still prove the classification. No failed route is counted as support for the successful theorem.

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